

Splunk Threat Hunting

Multi-Source SOC Investigation

Technical Project Report

A controlled SIEM investigation lab focused on Apache web logs, firewall telemetry, Windows authentication events, SPL-based threat hunting, cross-source correlation, timeline reconstruction, and SOC dashboarding.

Environment: VirtualBox-based controlled lab

Primary SIEM: Splunk

Telemetry: Apache access logs, simulated firewall logs, Windows Security logs

1. Project Overview

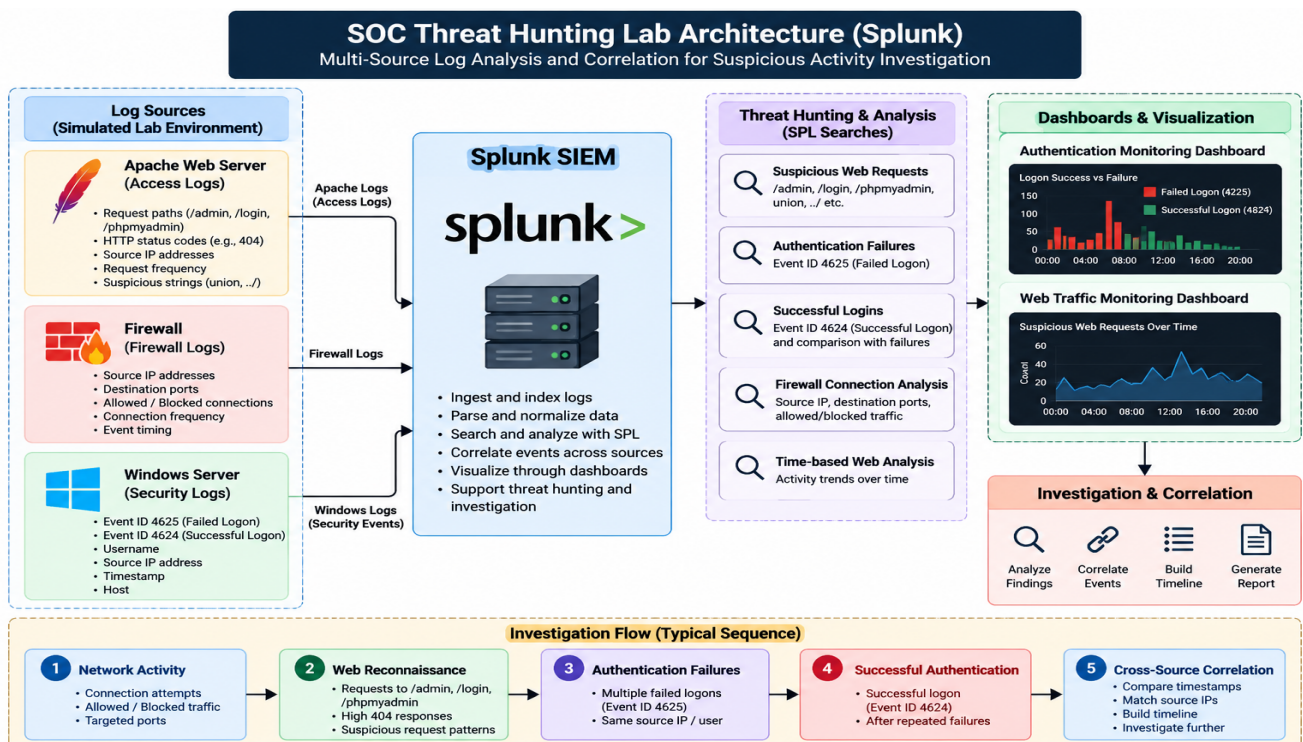
The project simulates a Tier-1 SOC analyst investigation in a controlled virtual environment. The objective was to examine multiple security telemetry sources, identify suspicious activity, and correlate events to reconstruct a plausible sequence of activity.

The project documentation describes Apache web-server logs, simulated firewall logs, and Windows Security logs as the principal data sources. Splunk was used for centralized search, analysis, correlation, and dashboard-based monitoring.

2. Objectives

- Analyze multiple security log sources in Splunk.
- Investigate suspicious web requests and reconnaissance behavior.
- Analyze failed and successful Windows authentication events.
- Examine firewall connection activity and targeted ports.
- Correlate source addresses and timestamps across telemetry.
- Perform time-based threat hunting.
- Use dashboards to support SOC monitoring and investigation.
- Document findings, limitations, and recommended defensive controls.

3. Architecture



The architecture separates telemetry collection from analysis. Apache, firewall, and Windows Security events are brought into Splunk, where SPL searches support threat hunting and correlation. The resulting analysis feeds monitoring dashboards and an investigation workflow.

4. Data Sources and Investigation Coverage

Apache Access Logs

The Apache investigation examined request paths, HTTP status codes, source addresses, request frequency, and suspicious patterns. Sensitive paths included /admin, /login, and /phpmyadmin. HTTP 404 responses were reviewed as potential indicators of resource probing.

Firewall Logs

Firewall telemetry was examined for repeated connection attempts, source IP addresses, target ports, event frequency, timing, and allowed versus blocked traffic.

Windows Security Logs

Event ID 4625 was used to identify failed logons and Event ID 4624 was used to identify successful logons. Authentication activity was evaluated together with source address, username, host, and timestamp where available.

5. SPL Threat-Hunting Searches

Suspicious web requests

```
index=* ("union" OR "../" OR "/admin" OR "/phpmyadmin")
| stats count by host, source, uri_path, status
| sort - count
```

Failed authentication

```
index=* (EventCode=4625 OR EventID=4625)
| stats count by host, src_ip, user
| sort - count
```

Successful authentication

```
index=* (EventCode=4624 OR EventID=4624)
| stats count by host, src_ip, user
| sort - count
```

6. Firewall and Time-Based Analysis

Firewall connection analysis

```
index=* (action=blocked OR action=allowed)
| stats count by src_ip, dest_port, action
| sort - count
```

Time-based suspicious web activity

```
index=* ("union" OR "../" OR "/admin" OR "/phpmyadmin")
| timechart span=1h count
```

7. Investigation Methodology

- 1. Establish hypothesis** — Start with an activity pattern that may indicate reconnaissance, suspicious web activity, or unauthorized authentication.
- 2. Analyze independently** — Review each telemetry source independently before making cross-source conclusions.
- 3. Identify suspicious patterns** — Use SPL searches to isolate sensitive paths, suspicious strings, authentication events, and repeated network activity.
- 4. Correlate events** — Compare timestamps and source addresses across Apache, firewall, and Windows telemetry.
- 5. Reconstruct timeline** — Arrange relevant events chronologically to understand the likely progression of observed activity.
- 6. Assess evidence** — Separate directly observed telemetry from correlation and analyst interpretation.
- 7. Recommend controls** — Identify defensive measures appropriate to the observed behavior.

8. Investigation Findings

- Repeated requests targeted sensitive web paths such as /admin, /login, and /phpmyadmin.
- High-frequency HTTP 404 responses were observed during web activity analysis.
- Suspicious request patterns including union and ../ were investigated.
- Multiple failed Windows authentication events were observed.
- Successful authentication events were examined in relation to failed authentication activity.
- Firewall telemetry was used to examine repeated connections, target ports, and allowed/blocked activity.
- Splunk dashboards provided visibility into authentication activity and web-traffic trends.

Assessment: The observations are consistent with suspicious reconnaissance and authentication activity in the controlled lab. They support further investigation but do not, by themselves, establish successful compromise of a production system.

9. SOC Dashboards

Authentication Monitoring

The authentication dashboard compares successful and failed Windows logon activity. It supports investigation of abnormal authentication patterns and provides context for source address, user, host, and timestamp analysis.

Web Traffic Monitoring

The web-traffic dashboard supports investigation of request paths, HTTP status codes, request frequency, suspicious strings, and changes in activity over time.

10. Recommended Defensive Controls

- Implement account lockout policies to reduce brute-force exposure.
- Enforce strong authentication and multi-factor authentication.
- Restrict access to administrative web directories.
- Use firewall controls to limit repeated suspicious connection attempts.
- Continuously monitor authentication logs and configure SIEM alerting.
- Keep systems patched and maintain appropriate security monitoring.

11. Limitations and Evidence Boundary

This project is a controlled cybersecurity lab. The project documentation states that log files were collected or simulated to represent realistic attack scenarios in a safe and repeatable environment. The repository therefore presents the activity as lab telemetry rather than as evidence of a real-world production compromise.

Temporal proximity and shared source addresses are treated as supporting evidence, not conclusive proof that all observed events were caused by the same actor. Suspicious request patterns are treated as investigation indicators rather than proof of successful exploitation.

12. Evidence Index

ID	Evidence	Purpose
E01	Apache logs in Splunk	Web telemetry ingestion and analysis
E02	Firewall logs in Splunk	Network activity analysis
E03	Windows Security events	Authentication telemetry
E04	Cross-log correlation	Multi-source investigation
E05	Suspicious web-request search	Threat hunting
E06	Time-based web analysis	Activity trend analysis
E07	Authentication dashboard	Success/failure monitoring
E08	Web-traffic dashboard	Web activity visualization

The public repository should contain only selected, non-sensitive evidence. Raw archives and screenshots containing credentials, tokens, or unnecessary personal data should remain private or be redacted before publication.

13. Project Outcome

The project demonstrates practical SIEM threat-hunting skills across multiple telemetry sources, including SPL searching, web-log analysis, authentication monitoring, firewall analysis, cross-source correlation, timeline reconstruction, and dashboard-based SOC monitoring.